

Dilatancy Test for Soil Explained

When the hand is bumped, the soil becomes very shiny, because water doesn't stick to silt particles and will come to the surface (**Figure 13-4**). This identifies the soil as silt, not clay. Severe bumping may even cause the soil to liquefy—turn to a fluid—also indicating that the soil is silt, not clay. Clays don't exhibit this behavior, because water sticks to clay particles and can't move readily.



Figure 13-4. a) squeezing; b) releasing (dull sheen); and c) vibrating silt in the hand (shiny).

Explanation of the Demonstration

When the silt was saturated, it collapsed into a dense state. When the silt was squeezed, it increased in volume, or dilated. That is, the void space between the soil particles increased. Although this is counterintuitive, it's the truth! Most things decrease in volume when squeezed, but not dense silts. As it dilated, water was sucked into the soil, leaving the surface devoid of water and the color of the silt—not shiny. Think of a squeezed saturated sponge; when it's released, it sucks up water, as did the silt. Now, when the hand was bumped, the soil collapsed, reducing the void volume and squeezing water out. The water, coming to the surface, produced a shiny appearance. If you have really pure silt, note carefully that the collapsing soil changes shape and begins to flow. This is called liquefaction (quicksand), which happens to saturated silt (and sand) deposits during earthquakes.

Source: adapted from Elton, D.J. 2015. Grounded! Amazing Classroom Demonstrations in Soil Mechanics. ASCE Press, Reston, VA.